The Charge Gap

Project compilation — a data investigation into Medicare hospital billing and the Medicaid coverage it sits alongside

PREPARED BY	Neha Vittal
COMPILED	October 26, 2026
CORE DATASET	CMS Medicare Provider Utilization & Payment Data, Inpatient, FY2014 (202,656 records)
CONTEXT DATASET	KFF Medicaid & CHIP Monthly Enrollment (most recent available month)

01

The problem, in simple terms

Hospital pricing in the U.S. is a black box. Hospitals set a "list price" for each procedure, but that price barely relates to what actually gets paid — so nobody (patients, employers, policymakers) can tell whether a hospital's pricing is reasonable, or which states and procedures carry outsized cost risk. That opacity is buried inside millions of individual billing records that no one actually reads.

02

The solution

This project takes CMS's real Medicare billing records — 202,656 of them — and turns that buried data into something anyone can see and compare at a glance: a set of charts and a map showing, nationally and state by state, exactly how far hospital "list prices" stray from what's actually paid, and where the biggest outliers are. Current Medicaid enrollment is placed alongside it, so the scale of both programs is visible side by side.

03

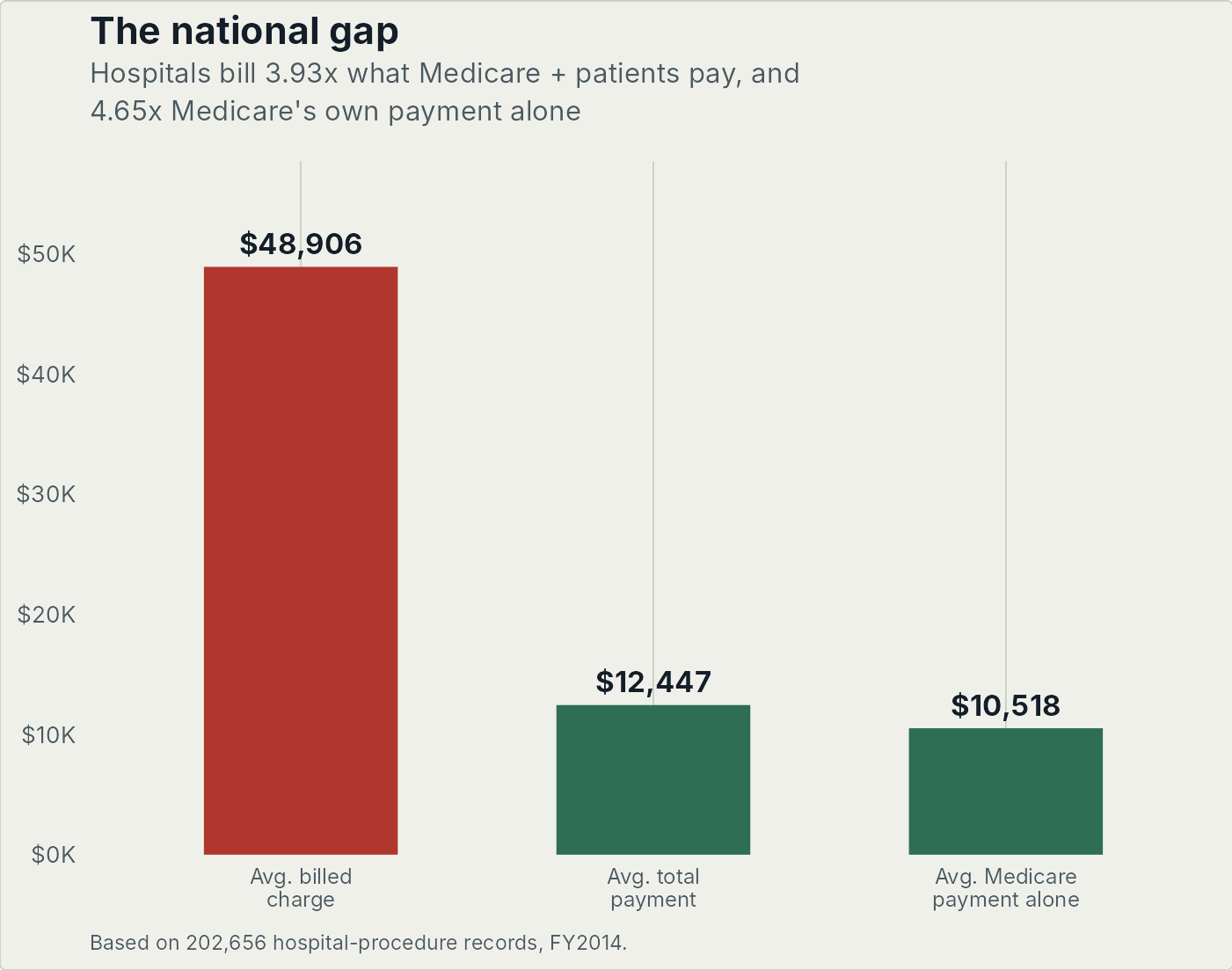
The national gap

Across every hospital-procedure record in the file, discharge-weighted:

AVG. BILLED CHARGE	AVG. TOTAL PAYMENT	AVG. MEDICARE PAYMENT ALONE

$48,906	$12,447	$10,518

Hospitals bill **3.93×** what Medicare and patients combined actually pay, and **4.65×** what Medicare pays on its own.

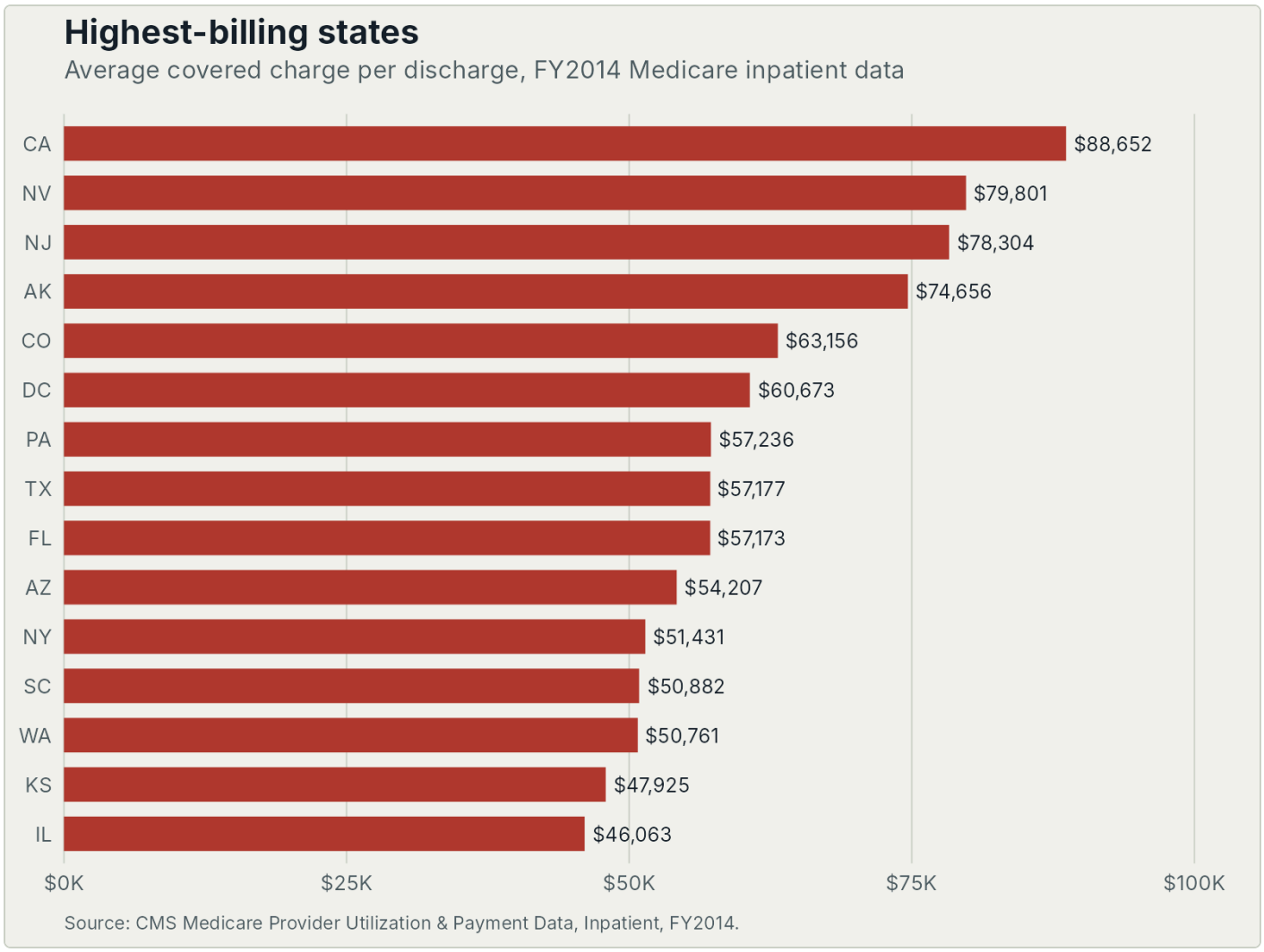


Built in R (ggplot2) from the full 202,656-record FY2014 file.

04

A geography of billing

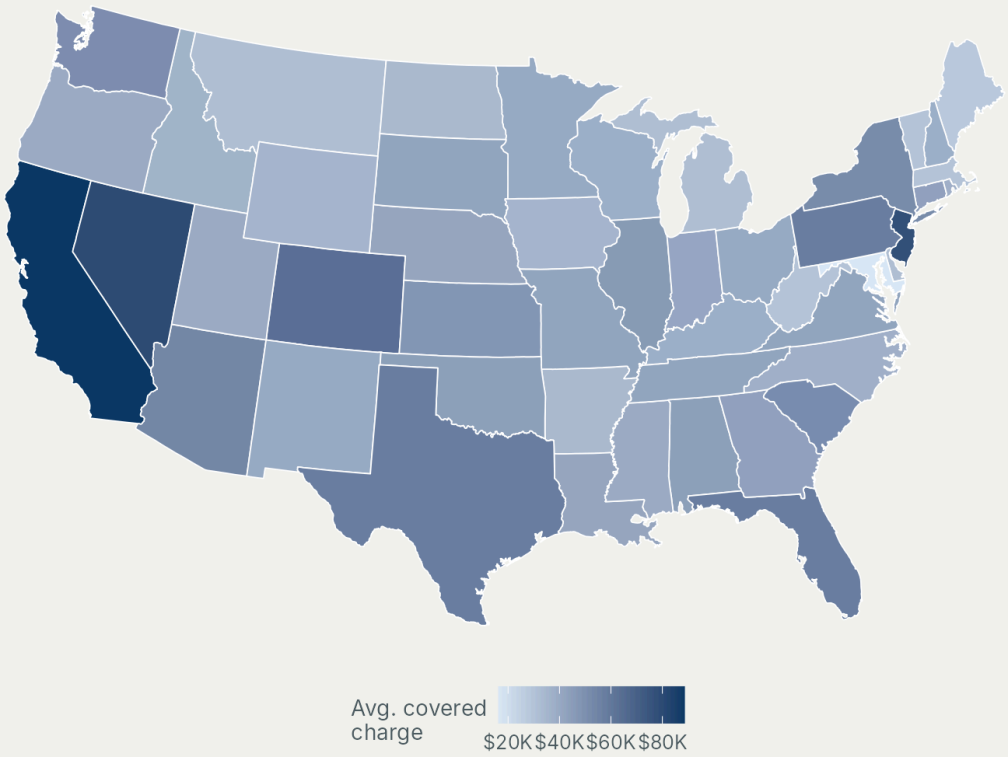
The gap isn't uniform. California hospitals bill **$88,652** on average — well over **5×** Maryland's **$16,286**, the lowest in the country. Maryland is a special case: it runs its own all-payer rate-setting system, so every payer, Medicare included, pays close to the same rate.



Highest-billing states by average covered charge per discharge.

A geography of billing

Average Medicare covered charge per discharge, by state (FY2014)



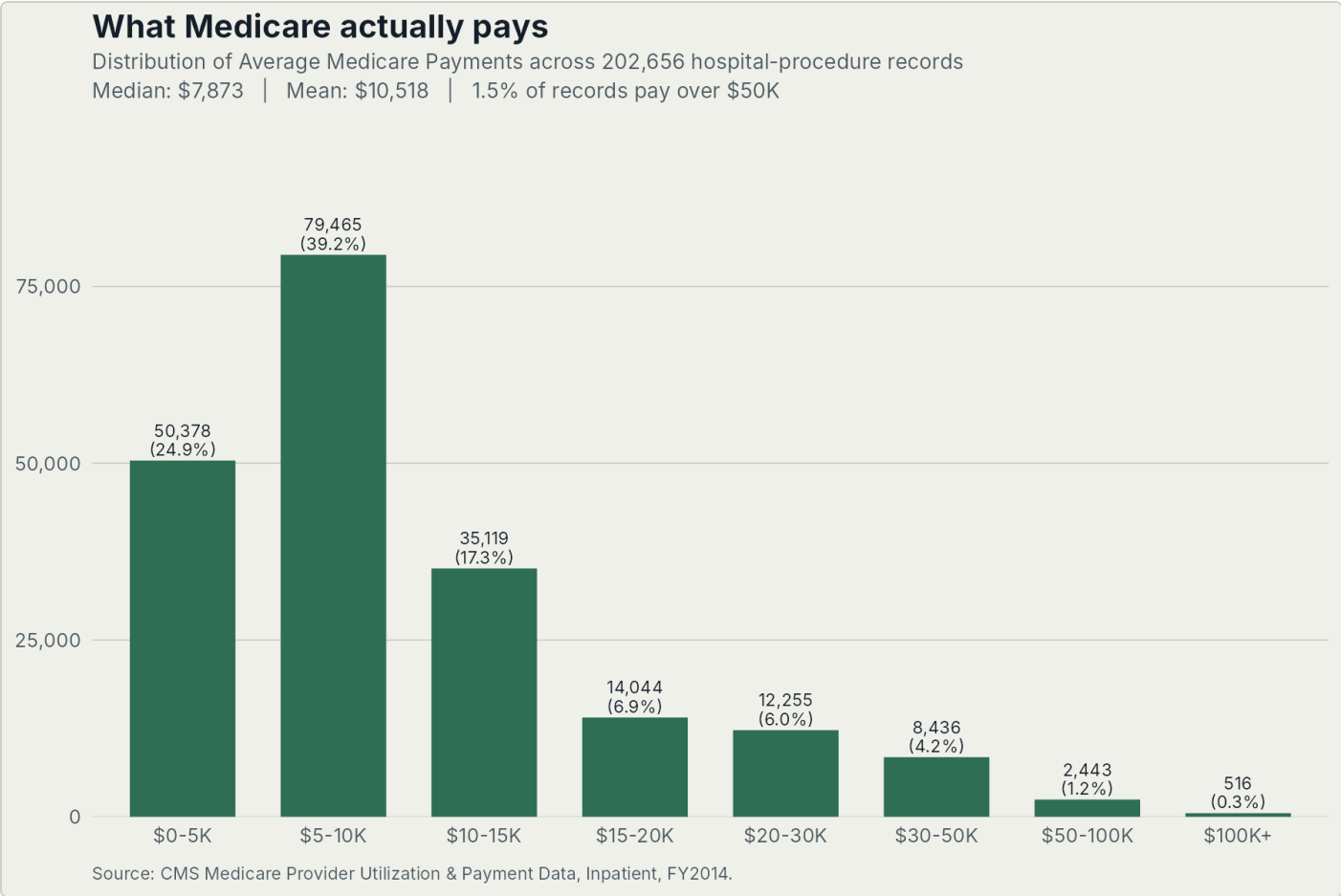
Source: CMS Medicare Provider Utilization & Payment Data, Inpatient, FY2014. Alaska and Hawaii not shown (continental-US geometry only).

State choropleth, continental U.S. (Alaska and Hawaii omitted — see Methodology).

05

What Medicare actually pays

Averages hide shape. Broken into cost bands, most Medicare inpatient payments are modest: the median payment is **$7,873**, well below the **$10,518** mean. Just **1.5%** of records pay more than $50,000 — but that thin tail is where transplants, extended ECMO stays, and severe burn treatment live.

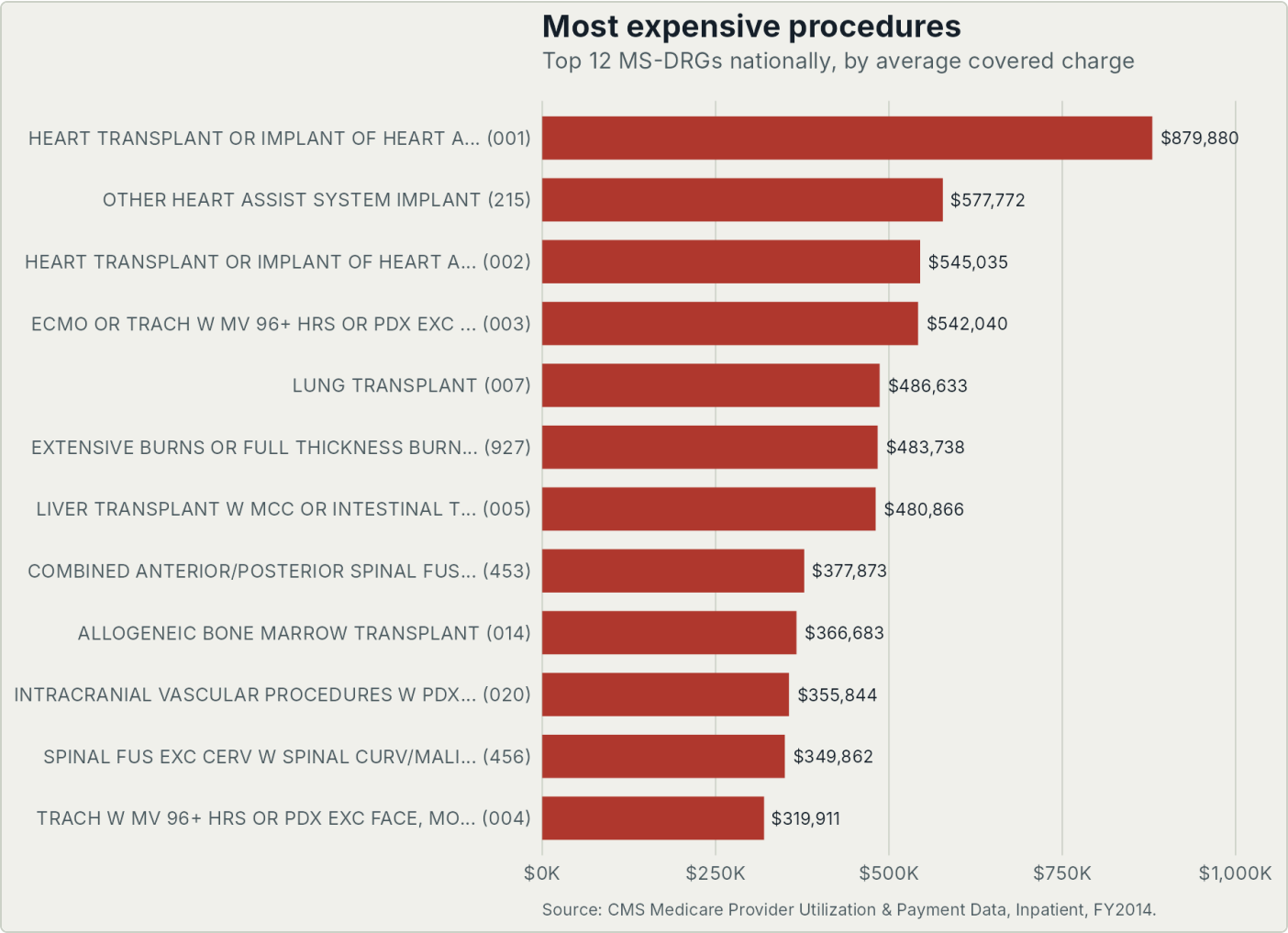


Distribution of Average Medicare Payments across all 202,656 records.

06

Where the tail leads

At the far end: a heart transplant billed at an average of **$879,880** — the single most expensive procedure in the file — followed by other heart-assist implants, ECMO with extended ventilation, and organ transplants generally.



Top 12 MS-DRGs nationally, by average covered charge.

07

Medicaid, for context

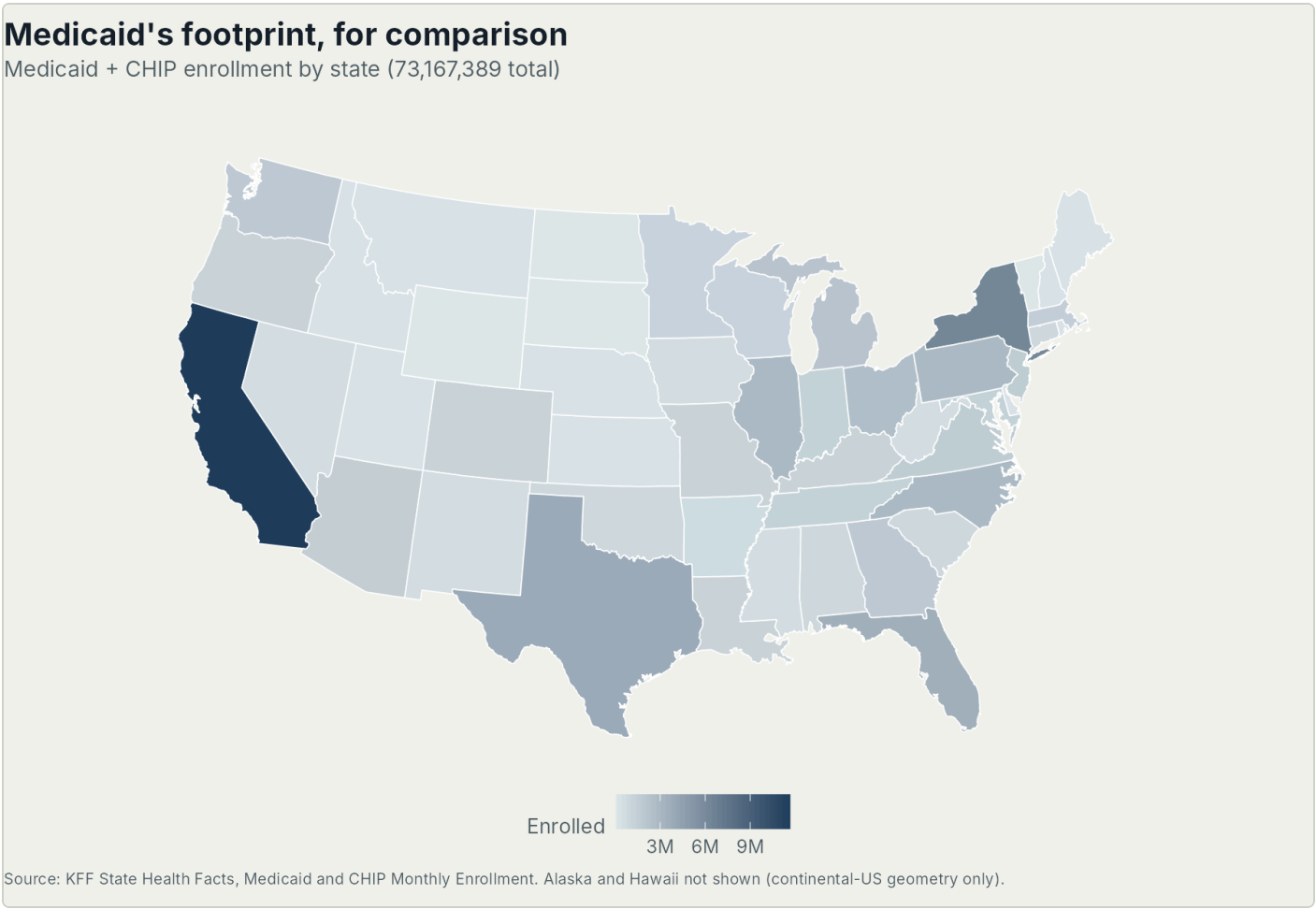
Medicaid doesn't publish a "billed vs. paid" figure the way Medicare does. What it does report, monthly, is how many people rely on it: **73,167,389** people enrolled in Medicaid and CHIP combined, as of the most recent KFF data.

MEDICAID + CHIP ENROLLED

73,167,389

HIGHEST-ENROLLMENT STATE

California



Medicaid + CHIP enrollment by state, continental U.S.

08

All deliverables produced in this project

README.md MARKDOWN

Project overview, problem statement, dataset documentation, and key findings for the GitHub repo.

Medicare Inpatient Charges Data Dictionary .DOCX

Full column-by-column documentation of the FY2014 source file, with summary statistics and limitations.

The Charge Gap — Case Study .DOCX

Narrative write-up: executive summary, methodology, four findings with embedded charts, limitations, and sources.

The Charge Gap — Deck .PPTX

10-slide presentation covering the same findings, built for a general portfolio audience.

The Charge Gap — Portfolio website `.HTML / LIVE ARTIFACT`

Single-page interactive site embedding all three original HTML/JS charts (national summary, payment distribution, state map toggle), plus a Medicaid coverage section.

Interactive charts `3× .HTML`

Self-contained HTML/SVG/JS charts with hover tooltips, light/dark mode, and table views — the charts embedded in the portfolio site above.

R chart scripts & renders `4× .R, 6× .PNG`

ggplot2 rebuild of the national summary, payment distribution, and state map charts — the source for every figure in this compilation.

> **A note on this compilation:** this document was assembled in a later session than the one that produced the README, data dictionary, case study, deck, and portfolio website. Those files were delivered earlier in this engagement and aren't in this session's storage to re-attach here, so this report documents and summarizes them rather than embedding their original content. The charts above are freshly rendered from the R scripts, which are in this session.

09

Methodology & limitations

- **Medicare fee-for-service only.** Excludes Medicare Advantage, the uninsured, and privately insured patients.
- **Averages, not itemized bills.** Each row is a per-hospital average across a year of claims for one procedure, not an individual patient's bill.
- **Maryland is structurally different.** Its all-payer rate system means its low ratio reflects policy, not lower costs.
- **Small-volume records suppressed.** Hospital-procedure pairs with fewer than 11 discharges/year are dropped by CMS.
- **One fiscal year, chosen deliberately.** CMS has published newer inpatient files since FY2014, but this project stays on FY2014 because it's the vintage available here as a complete, row-level file every figure can be traced back to.
- **Alaska and Hawaii omitted from the R maps.** The R package set available in this session only ships continental-U.S. polygon geometry; the original HTML/JS map (built with d3-geo) includes both as insets.
- **Figures recomputed in a fresh session differ slightly from earlier deliverables.** An independent recomputation from the raw CMS file gave a national average charge of $48,906 (discharge-weighted), versus $52,589 used in the originally delivered deck, case study, and website. Both used the same methodology (discharge-weighted average); the discrepancy's exact source wasn't tracked down, since the original session's files weren't available to compare against directly. Every figure in this document was independently verified against the raw file before use.
